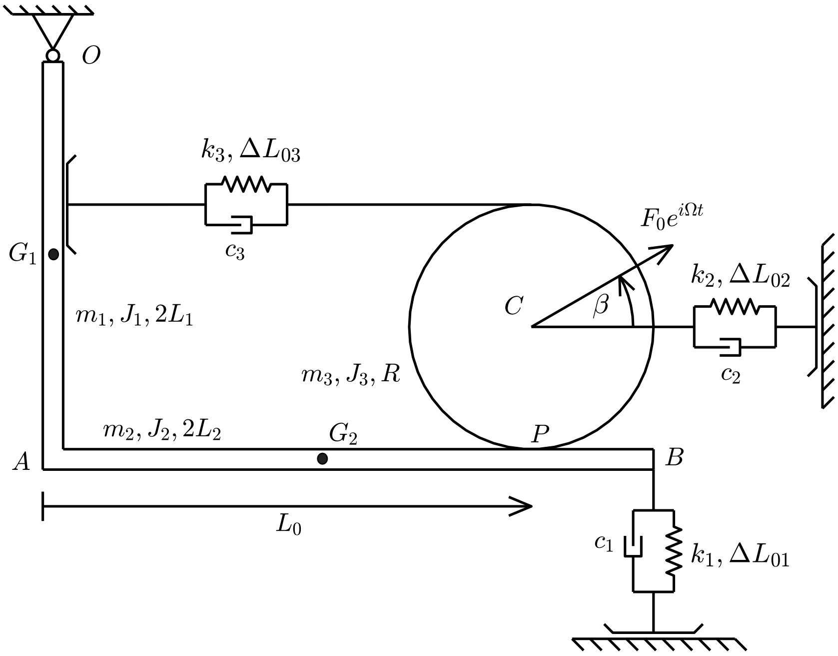
Dynamics of Mechanical Systems

August 31st, 2022

Problem 1:



$m_1 =$ 10 kg	$k_1 =$ 10000 N/m
$m_2 =$ 15 kg	$k_2 =$ 5000 N/m
$m_3 =$ 5 kg	$k_3 =$ 2500 N/m
$J_1 =$ 0.8 kgm^2	$\Delta L_{01} =$ -0.008 m
$J_2 =$ 2.5 kgm^2	$\Delta L_{02} =$ 0.01 m
$J_3 =$ 0.1 kgm^2	$\Delta L_{03} =$ 0.01 m
$L_1 =$ 0.5 m	$c_1 =$ 10.0 Ns/m
$L_2 =$ 0.75 m	$c_2 =$ 5.0 Ns/m
$L_0 =$ 1.0 m	$c_3 =$ 1.0 Ns/m
$R =$ 0.2 m	$\beta = \pi/6$

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The disk rolls without sliding on the horizontal side of the L-shaped bar.

The students are requested to:

0. Write on top of each page their surname, name, Person Code and **SNxxxx code** (see below)
1. Write the equations of motion of the system linearized around the given equilibrium configuration (provide this answer on paper: Jacobian matrices and second-order stiffness terms)
2. Compute the natural frequencies and modal shapes (provide this answer on paper, copying the values from MATLAB);
3. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic inclined force $F_0 e^{i\Omega t}$. Output: the horizontal and the vertical component of the displacement of point C. Save the graphical results as SNxxxx1.fig and SNxxxx2.fig, with SNxxxx your identifier, see below;
4. Compute the Frequency Response Function (in the range 0 – 10 Hz, step = 0.01 Hz) applying the harmonic inclined force $F_0 e^{i\Omega t}$. Output: the normal and tangential components of the contact forces between the L-shaped bar and the disk at point P. Save the graphical result as SNxxxx3.fig and SNxxxx4.fig.
5. Compute the static preloads ΔL_{01} and ΔL_{02}, keeping the same value of ΔL_{03}, when $m_3 = 10$ kg (provide the answer on paper).

The answer to points 1, 2 and 5 shall be provided on paper (for point 2 please copy on paper the values of the natural frequencies and eigenmodes produced by the MATLAB script); the answer to points 3 – 4 shall be provided by delivering the MATLAB script and .fig files as a compressed archive uploaded on WeBeeP, see instructions below.

You must upload one single archive file for both problem 1 and problem 2, which should contain your MATLAB scripts and all the .fig files you want to deliver as part of your solution.

i.) Accepted formats for the archive files are: .7z .rar .zip;
ii.) The file must be named SNxxxx with "S" the first letter in your surname, "N" the first letter in your name (in case of multiple surnames/names just consider the first one), "xxxx" the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the name of the file should be: BS5678.ext;
iii.) The file must be uploaded as an assignment on WeBeeP, the name of the assignment is **Test_31_August_2022**;
iv.) the deadline to upload the assignment will be announced during the written exam.

Your identifier: SNxxxx with S the first letter of your surname, N the first letter of your name and xxxx the last four digits of your Person Code (Codice Persona), e.g. for Bruni Stefano 12345678 the identifier BS5678

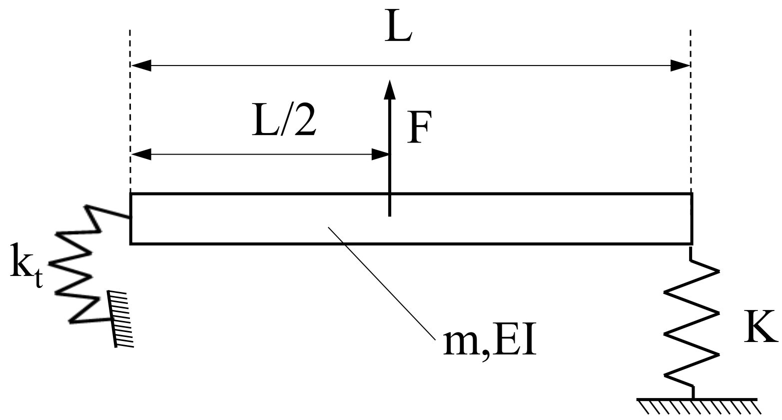
Please be aware that your answers to the test will not be considered in case:

- **a. the information stated at point 0 above is missing;**
- **b. your archive file is uploaded in the wrong folder**
- **c. the name of the archive file is different from the identifier SNxxxx described above;**

For environmental reasons the students are kindly requested to use all of the four pages of each sheet before using another one.

Problem 2:

Consider the <u>bending motion</u> of a beam with uniform section and material properties having length L. The left end of the beam is connected to a torsional spring having stiffness k_t and the opposite end is supported by a linear spring having stiffness k.



DATA:
EI=2e5 Nm^2
m=5 kg/m
L=2 m
k_t = 1e5 Nm
k = 5e6 N/m

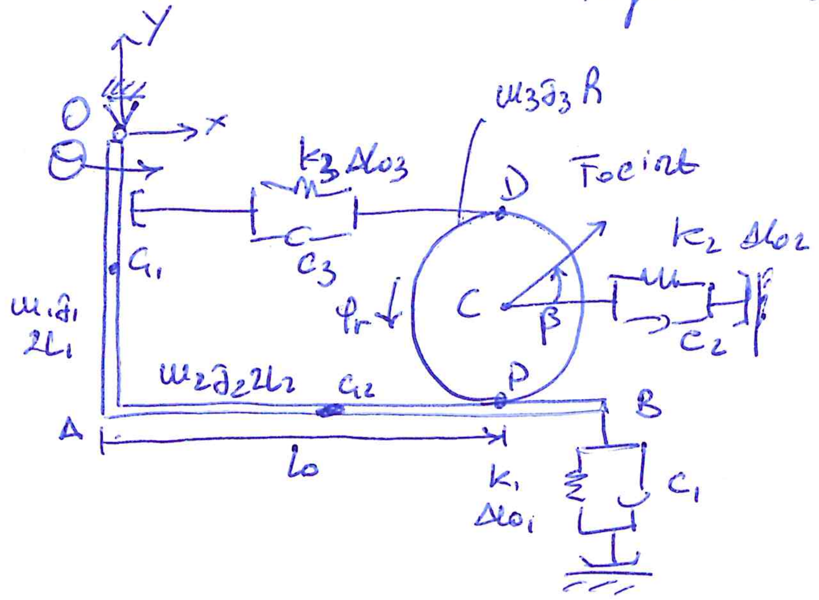
1) Write the boundary conditions and the expression of the square matrix of coefficients H of the linear homogeneous system defining the natural frequencies of the system. Deliver this part of the solution on paper.
2) Compute the natural frequencies of the system in the frequency range 0-200 Hz and the related modal shapes. Deliver the values of the natural frequencies on paper and include in your archive file one .fig file for each modal shape. The .fig files shall be named SNxxxxP2i with SNxxxx your identifier (see below) and i=1,2,… the number of the mode (use as many as needed);
3) Plot the Bode diagram (log y scale) of the magnitude of the FRF considering:
   - Non-dimensional damping factor: 0.01 for all modes
   - input: force having unit magnitude applied in the centre of beam as shown in the figure;
   - output: vertical vibration in the right end of the beam;

   The Bode diagram shall be delivered as a .fig file named SNxxxxP2FRF included in your archive.

<u>The answer to points 1 shall be provided on paper; the answer to points 2 – 3 shall be provided by delivering **the MATLAB script** and .fig files as a compressed archive uploaded on WeBeeP, see instructions reported for problem 1.</u>

DYNAMICS OF MECHANICAL SYSTEMS

Written Test August 31^{st} 2022



- 2 rigid bodies (6 dof)
- 1 cart and rolling without sliding cond. → 4 dofc.

→ 2 dof.

$$\underline{x} = \begin{Bmatrix} \theta \\ \varphi_r \end{Bmatrix} \qquad \underline{x}_0 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

φ_r : relative rotation of the disk with respect to the first rigid body

$$T = \frac{1}{2} m_1 \dot{x}_{G_1}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{G_2}^2 + \frac{1}{2} m_2 \dot{y}_{G_2}^2 + \frac{1}{2} J_2 \dot{\theta}^2 + \frac{1}{2} m_3 \dot{x}_C^2 + \frac{1}{2} m_3 \dot{y}_C^2 + \frac{1}{2} J_3 \dot{\varphi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^{3} c_i\, \dot{\Delta l}_i^2 \ ; \quad V_{el} = \frac{1}{2} \sum_{i=1}^{3} k_i\, \Delta l_i^2 \ ; \quad V_g = \sum_{i=1}^{3} m_i g\, y_{G_i}$$

$$\delta W = \vec{F}_0 e^{i\Omega t} \times \delta \vec{S}_C = F_0 \cos\beta\, e^{i\Omega t} \delta x_C + F_0 \operatorname{sen}\beta\, e^{i\Omega t} \delta y_C$$

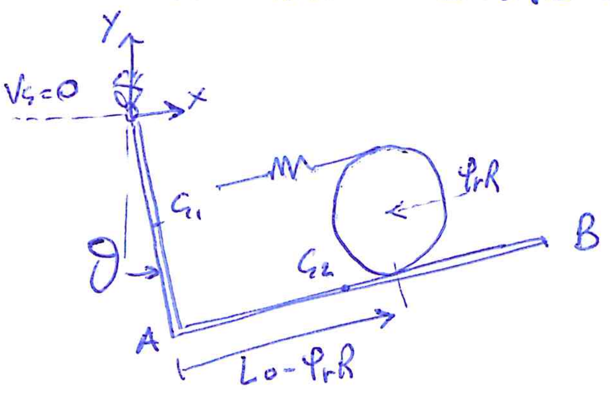
$$[M_{phy}] = diag(m_1\ J_1\ m_2\ m_2\ J_2\ m_3\ m_3\ J_3)$$

$$\underline{y}_M = \{ x_{G_1}\ \theta\ x_{G_2}\ y_{G_2}\ \theta\ x_C\ y_C\ \varphi \}^T$$

$$[C_{phy}] = diag(c_1\ c_2\ c_3) \ ; \ [k_{phy}] = diag(k_1\ k_2\ k_3) \ ; \ \underline{y}_c = \underline{y}_k = \{ \Delta l_1\ \Delta l_2\ \Delta l_3 \}^T$$

$$\underline{F} = \begin{Bmatrix} F_0 \cos\beta \\ F_0 \operatorname{sen}\beta \end{Bmatrix} e^{i\Omega t} \qquad \underline{y}_e = \begin{Bmatrix} x_C \\ y_C \end{Bmatrix}$$

- non linear kinematic relationships



$$x_{G_2} = 2L_1 \operatorname{sen}\vartheta + L_2 \cos\vartheta$$

$$y_{G_2} = -2L_1 \cos\vartheta + L_2 \operatorname{sen}\vartheta$$

$$x_C = 2L_1 \operatorname{sen}\vartheta + (L_0 - \varphi_r R)\cos\vartheta - R\operatorname{sen}\vartheta$$

$$y_C = -2L_1 \cos\vartheta + (L_0 - \varphi_r R)\operatorname{sen}\vartheta + R\cos\vartheta$$

$$y_B = -2L_1 \cos\vartheta + 2L_2 \operatorname{sen}\vartheta$$

$$\Delta l_1 = \Delta y_B = y_B - y_{B0}$$

$$\Delta l_3 = -2\varphi_r R \quad \text{(P is the relative I.A.R. of the disk)}$$

$$\Delta l_2 = -\Delta x_C = -x_C + x_{C0}$$

$$x_{G1,lin} = L_1 \vartheta \quad ; \quad \varphi = \vartheta + \varphi_r$$

$$\frac{\partial x_{G2}}{\partial \vartheta} = 2L_1 \cos\vartheta - L_2 \operatorname{sen}\vartheta$$

$$\frac{\partial y_{G2}}{\partial \vartheta} = 2L_1 \operatorname{sen}\vartheta + L_2 \cos\vartheta$$

$$\frac{\partial x_c}{\partial \vartheta} = (2L_1 - R)\cos\vartheta - (L_0 - R\varphi_r)\operatorname{sen}\vartheta \quad ; \quad \frac{\partial x_c}{\partial \varphi_r} = -R\cos\vartheta$$

$$\frac{\partial y_c}{\partial \vartheta} = (2L_1 - R)\operatorname{sen}\vartheta + (L_0 - R\varphi_r)\cos\vartheta \quad ; \quad \frac{\partial y_c}{\partial \varphi_r} = -R\operatorname{sen}\vartheta$$

$$\frac{\partial \Delta L_2}{\partial \vartheta} = -(2L_1 - R)\cos\vartheta + (L_0 - R\varphi_r)\operatorname{sen}\vartheta \quad ; \quad \frac{\partial \Delta L_2}{\partial \varphi} = R\cos\vartheta$$

$$\frac{\partial \Delta L_1}{\partial \vartheta} = 2L_1 \operatorname{sen}\vartheta + 2L_2 \cos\vartheta$$

$$[\Lambda_u] = \begin{bmatrix} L_1 & 0 \\ 1 & 0 \\ 2L_1 & 0 \\ L_2 & 0 \\ 1 & 0 \\ 2L_1 - R & -R \\ L_0 & 0 \\ 1 & 1 \end{bmatrix} \qquad [\Lambda_{\delta k}] = [\Lambda_c] = \begin{bmatrix} 2L_2 & 0 \\ -(2L_1 - R) & R \\ 0 & -2R \end{bmatrix}$$

$$[\Lambda_e] = \begin{bmatrix} 2L_1 - R & -R \\ L_0 & 0 \end{bmatrix}$$

$$h_{G1} = y_{G1} \quad ; \quad h_{G2} = y_{G2} \quad ; \quad h_{G3} = y_c$$

$$\frac{\partial^2 h_{G1}}{\partial \vartheta^2} = L_1 \cos\vartheta \qquad [k_{G1}] = m_1 g \begin{bmatrix} L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 h_{G2}}{\partial \vartheta^2} = 2L_1 \cos\vartheta - L_2 \operatorname{sen}\vartheta \qquad [k_{G2}] = m_2 g \begin{bmatrix} 2L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial^2 h_{G3}}{\partial \vartheta^2} = (2L_1 - R)\cos\vartheta - (L_0 - R\varphi_r)\operatorname{sen}\vartheta \\[2ex] \dfrac{\partial^2 h_{G3}}{\partial \vartheta \partial \varphi_r} = -R\cos\vartheta \quad ; \quad \dfrac{\partial^2 h_{G3}}{\partial \varphi_r^2} = 0 \end{cases} \qquad k_{G3} = m_3 g \begin{bmatrix} 2L_1 - R & -R \\ -R & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta l_1}{\partial\theta^2} = 2L_1\cos\vartheta - 2L_2\,\mathrm{sen}\,\vartheta \qquad [K_{el\,II\,1}] = k_1\Delta l_{01}\begin{bmatrix} 2L_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial^2 \Delta l_2}{\partial\theta^2} = (2L_1 - R)\,\mathrm{sen}\,\vartheta + (L_0 - R\varphi_r)\cos\vartheta \\[2ex] \dfrac{\partial^2 \Delta l_2}{\partial\theta\,\partial\varphi_r} = -R\,\mathrm{sen}\,\vartheta \quad ; \quad \dfrac{\partial^2 \Delta l_2}{\partial\varphi_r^2} = 0 \end{cases} \qquad [K_{el\,II,2}] = k_2\Delta l_{02}\begin{bmatrix} L_0 & 0 \\ 0 & 0 \end{bmatrix}$$

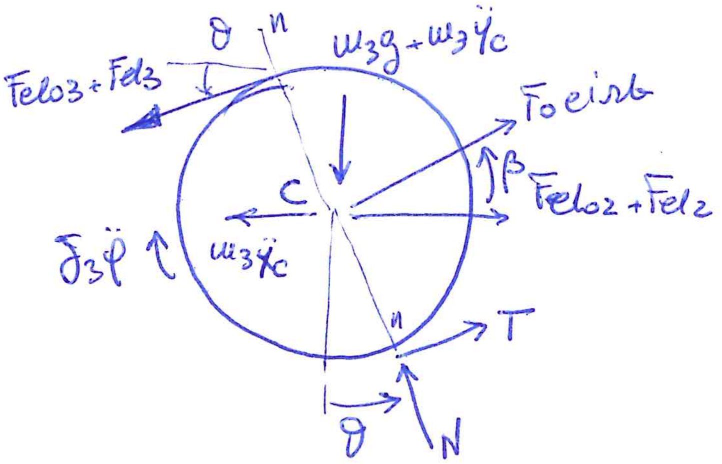
- FRF

input: $F_0 e^{i\Omega t}$

outputs: $x_c \simeq (2L_1 - R)\,\theta - R\varphi_r$

$y_c \simeq L_0\,\theta$

N_{dyn}, T_{dyn} :



$\sum M_c^{DISK} = 0$:

$$TR + (F_{el03} + F_{el3})R - J_3\ddot{\varphi} = 0$$

$$T_{ST} = -F_{el03}$$

$$T_{dyn} = \frac{J_3\ddot{\varphi}}{R} - F_{el3}$$

$$\sum F_{n\text{-}n}^{DISK} = 0 : \quad N + m_3\ddot{x}_c\,\mathrm{sen}\,\vartheta - (F_{el02} + F_{el2})\,\mathrm{sen}\,\vartheta - F_0\cos\beta\, e^{i\Omega t}\,\mathrm{sen}\,\vartheta + $$
$$+ F_0\,\mathrm{sen}\,\beta\, e^{i\Omega t}\cos\vartheta - (m_3 g + m_3\ddot{y}_c)\cos\vartheta = 0$$

$$N + m_3\ddot{x}_c\,\mathrm{sen}\,\vartheta_0 + (\cancel{m_3\ddot{x}_c\cos\vartheta_0})\,\theta - F_{el02}\,\mathrm{sen}\,\vartheta_0 - (F_{el02}\cos\vartheta_0)\,\theta +$$
$$- F_{el2}\,\mathrm{sen}\,\vartheta_0 - (\cancel{F_{el2}\cos\vartheta_0})\,\theta - F_0\cos\beta\, e^{i\Omega t}\,\mathrm{sen}\,\theta_0 + F_0\,\mathrm{sen}\,\beta\, e^{i\Omega t}\cos\vartheta_0 +$$
$$- m_3 g\cos\vartheta_0 + (m_3 g\,\mathrm{sen}\,\vartheta_0)\,\theta - m_3\ddot{y}_c\cos\vartheta_0 + (\cancel{m_3\ddot{y}_c\,\mathrm{sen}\,\vartheta_0})\,\vartheta = 0$$

$$N_{ST} = F_{el02}\,\mathrm{sen}\,\vartheta_0 + m_3 g\cos\vartheta_0$$

$$N_{dyn} = -m_3\ddot{x}_c\,\mathrm{sen}\,\vartheta_0 + F_{el_2}\,\mathrm{sen}\,\vartheta_0 + (F_0\cos\beta\,\mathrm{sen}\,\vartheta_0 - F_0\,\mathrm{sen}\,\beta\cos\vartheta_0)\,e^{i\Omega t} +$$
$$+ m_3\ddot{y}_c\cos\vartheta_0 + \underbrace{(F_{el02}\cos\vartheta_0 - m_3 g\,\mathrm{sen}\,\vartheta_0)\,\vartheta}_{\text{additional non-mandatory linear Terms}}$$

$$\ddot{\varphi} = -\Omega^2 \varphi = -\Omega^2(\theta + \varphi_r)$$

$$\ddot{x}_C = -\Omega^2 x_C \; ; \; \ddot{y}_C = -\Omega^2 y_C$$

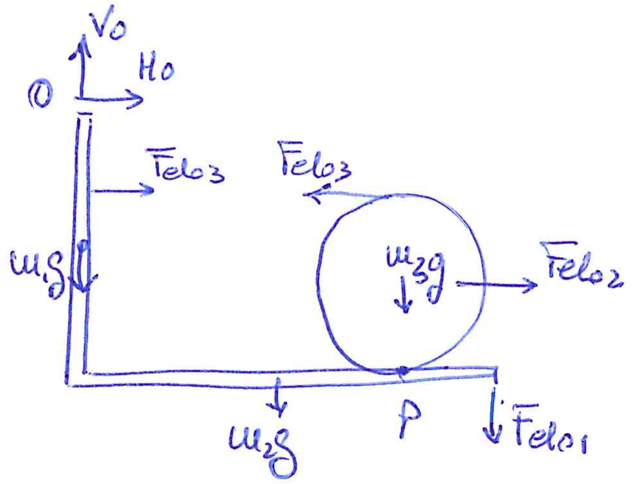
$$F_{el3} = (k_3 + i\Omega C_3)\,\Delta l_3 \; ; \; \Delta l_3 \simeq -2R\varphi_r$$

$$F_{el2} = (k_2 + i\Omega C_2)\,\Delta l_2 \; ; \; \Delta l_2 \simeq -(2L_1 - R)\theta + R\varphi_r$$

being $\theta_0 = 0 \rightarrow \sin\theta_0 = 0$; $\cos\theta_0 = 1$

the 2nd order terms have been simplified.

- static preload of the springs:



$$\Sigma M_P^{DISK} = 0 : \; F_{elo3}\,2R - F_{elo2}\,R = 0$$

$$F_{elo2} = 2F_{elo3}$$

$$\Sigma M_O = 0 :$$

$$m_2 g L_2 + F_{elo1}\,2L_2 + m_3 g L_0 + - F_{elo2}(2L_1 - R) = 0$$

$$F_{elo1} = \frac{F_{elo2}(2L_1 - R) - m_2 g L_2 - m_3 g L_0}{2L_2}$$